

div / curl / grad / ∇??
→ Operators (Col 1)

Helix r(t)?
→ FS Chain (Col 1 btm)

Surface area?
→ Monge (Col 2 btm)

∫F·dr or work?
→ Line Int (Col 2–3)

∭ or volume?
→ Triple Int (Col 2)

∮ circulation?
→ Circ (Col 3)

SCOPE · PRIORITY · EXAM ORDER

In scope: triple integrals, cyl/sph setup, curves, helix/FS, surface area, grad/div/curl/Laplacian, line integrals, circulation.

1. Operators2. Helix / FS3. Surface area4. Line integrals5. Triple integrals

Weak-student rule: do not spread effort evenly. Make operator identities, helix template, Monge formula, and line-integral workflow nearly automatic first.

CORE OPERATORS · CARTESIAN

$\nabla = \mathbf{i} \partial_x + \mathbf{j} \partial_y + \mathbf{k} \partial_z$
 $\nabla f = (f_x, f_y, f_z)$ → vector output
 $\nabla \cdot \mathbf{F} = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$ → scalar output

$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}$ → middle term sign flips

$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}$ → sum all pure 2nd derivs
Always: $\nabla \times (\nabla f) = \mathbf{0}$, $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.

Product rules: $\nabla(uv) = u \nabla v + v \nabla u$, $\nabla \cdot (u\mathbf{F}) = u \nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla u$, $\nabla \times (u\mathbf{F}) = \nabla u \times \mathbf{F} + u \nabla \times \mathbf{F}$
Cross-product rule: $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$

If you see → scalar input: use ∇ or ∇^2 . Vector input: use $\nabla \cdot$ or $\nabla \times$. Check output type matches.

RADIAL IDENTITIES · MUST MEMORIZE

$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$
 $\nabla \cdot \mathbf{r} = 3$ → instant, $\nabla \times \mathbf{r} = \mathbf{0}$ → instant, $\nabla(r^2) = 2\mathbf{r}$ → instant
 $\nabla(r^n) = nr^{n-2}\mathbf{r} = nr^{n-1}\hat{\mathbf{r}}$, $\nabla(1/r) = -\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$
Radial Laplacian: $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$ → critical for $\ln r$, r^n types

If you see $(x^2 + y^2 + z^2)^n$ or $h(r)$ → convert to radial form $r^k \hat{\mathbf{r}}$ FIRST, then apply radial formula.

2024 auto-marks: $\nabla \cdot \mathbf{r} = 3$, $\nabla \times \mathbf{r} = \mathbf{0}$, $\nabla r^2 = 2\mathbf{r}$. Write these instantly.

Common trap: mixing $\mathbf{r}$ and r^2 , or writing scalar when vector is needed → fix: check output type after every computation.

CURVES · ARCLENGTH · HELIX

$\mathbf{r}(t) = (x(t), y(t), z(t))$, $\mathbf{v} = \mathbf{r}'(t)$, $ds = |\mathbf{r}'(t)| dt$, $s(t) = \int_{t_0}^t |\mathbf{r}'(u)| du$

Helix: $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$; $\mathbf{r}' = (-a \sin t, a \cos t, b)$; $|\mathbf{r}'| = c = \sqrt{a^2 + b^2}$ → constant

If you see $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$ → it is a helix. Execute the FS chain below.

Exam sequence: differentiate → speed → arclength → unit tangent → unit normal → binormal → curvature/torsion.

FRENET-SERRET CORE

$\hat{\mathbf{t}} = \frac{d\mathbf{r}}{ds} = \frac{\mathbf{r}'}{|\mathbf{r}'|}$ → normalize first

$\kappa = \left| \frac{d\hat{\mathbf{t}}}{ds} \right|$

$\frac{d\hat{\mathbf{t}}}{ds} = \kappa \hat{\mathbf{n}}$ → then $\hat{\mathbf{n}} = \frac{1}{\kappa} \frac{d\hat{\mathbf{t}}}{ds}$

$\hat{\mathbf{b}} = \hat{\mathbf{t}} \times \hat{\mathbf{n}}$ → cross product order matters

$\frac{d\hat{\mathbf{b}}}{ds} = -\tau \hat{\mathbf{n}}$

Helix results: $\kappa = \frac{a}{a^2 + b^2}$, $\tau = \frac{b}{a^2 + b^2}$, $\Omega = \kappa \hat{\mathbf{b}} + \tau \hat{\mathbf{t}}$

Common trap: computing $d\hat{\mathbf{t}}/ds$ and calling it κ → fix: ALWAYS divide by $|\mathbf{r}'|$ to convert $d\mathbf{r} \rightarrow ds$

CYLINDRICAL / SPHERICAL OPERATORS

Cyl: $\nabla f = f_r \hat{\mathbf{r}} + \frac{1}{r} f_\theta \hat{\boldsymbol{\theta}} + f_z \hat{\mathbf{z}}$
 $\nabla \cdot \mathbf{F} = \frac{1}{r} \partial_r(r F_r) + \frac{1}{r} \partial_\theta F_\theta + \partial_z F_z$ → note $\frac{1}{r} \partial_r(r F_r)$ structure
 $\nabla^2 f = \frac{1}{r} \partial_r(r f_r) + \frac{1}{r^2} f_{\theta\theta} + f_{zz}$

Sph: $\nabla f = f_r \hat{\mathbf{r}} + \frac{1}{r} f_\phi \hat{\boldsymbol{\phi}} + \frac{1}{r \sin \phi} f_\theta \hat{\boldsymbol{\theta}}$
 $\nabla \cdot \mathbf{F} = \frac{1}{r^2} \partial_r(r^2 F_r) + \frac{1}{r \sin \phi} \partial_\phi(\sin \phi F_\phi) + \frac{1}{r \sin \phi} \partial_\theta F_\theta$
 $\nabla^2 f = \frac{1}{r^2} \partial_r(r^2 f_r) + \frac{1}{r^2 \sin \phi} \partial_\phi(\sin \phi f_\phi) + \frac{1}{r^2 \sin^2 \phi} f_{\theta\theta}$ → use if f depends on ϕ or θ

Purely radial $\mathbf{F} = F_r(r) \hat{\mathbf{r}}$: $\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d}{dr}(r^2 F_r)$; $\nabla \times \mathbf{F} = \mathbf{0}$ away from singularities.

If you see $(x^2 + y^2 + z^2)^n(x, y, z)$ → rewrite as $r^{2n+1} \hat{\mathbf{r}}$, identify F_r , use spherical div formula.

Common trap: forgetting $\hat{\mathbf{r}} = \mathbf{r}/r$ introduces extra r → fix: write F_r explicitly before applying formula.

COORDINATES · JACOBIANS · BOUNDS

Polar / Cyl: $x = r \cos \theta$, $y = r \sin \theta$, $dA = r dr d\theta$, $dV = r dr d\theta dz$
Spherical: $x = r \sin \phi \cos \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \phi$
 $dV = r^2 \sin \phi dr d\phi d\theta$ → never drop $\sin \phi$

First octant $\Rightarrow 0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$
Cone $z = r \Rightarrow \phi = \pi/4$
Sphere radius $a \Rightarrow r = a$

Offset cylinder $x^2 + (y-1)^2 = 1 \Rightarrow r = 2 \sin \theta$

Setup rule: sketch the shadow region first, then choose coordinates by the geometry trigger table.

TRIPLE-INTEGRAL MICRO-TEMPLATES

Volume: $V(D) = \iiint_D 1 dV$ → integrand is 1, not the surface
Centroid: $\bar{\mathbf{z}} = \frac{1}{V} \iiint_D z dV$ → divide by V at the end

Cartesian: inner variable runs from lower surface to upper surface. → sketch region first
Cylindrical: sidewall → r bound; floor → z lower; roof → z upper; θ from geometry. Attach Jacobian r .
Spherical: sphere → r bound; cone → ϕ bound; octant → θ bound. Attach $r^2 \sin \phi$

If you see $\iiint$ → sketch solid, pick coordinates by geometry, write bounds, attach Jacobian.

EP2 Q11 pattern: sidewall $r = 2 \sin \theta$, floor $z = 0$, roof $z = r$, $dV = r dz dr d\theta$

Common trap: writing $dV = dr d\theta dz$ without Jacobian → fix: cyl $\rightarrow r$, sph $\rightarrow r^2 \sin \phi$

SURFACE PARAMETRIZATION · AREA

Parametric surface: $\mathbf{X}(u, v)$
 $dS = |\mathbf{X}_u \times \mathbf{X}_v| du dv$
Graph surface: $z = f(x, y)$
 $\mathbf{X}(x, y) = \langle x, y, f(x, y) \rangle$
 $dS = \sqrt{1 + f_x^2 + f_y^2} dx dy$ → do not forget the $\sqrt{}$

Always find the projection region in the xy -plane before integrating. → sketch region

Paraboloid cap template: solve intersection first; if $z = 2 - r^2$ and cone $z = r$, then $2 - r^2 = r \Rightarrow r = 1$.

Then $A = \int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} r dr d\theta$

If you see "find the area of $z = f(x, y)$ " → Monge formula: $\sqrt{1 + f_x^2 + f_y^2}$ project onto xy -plane.

Common trap: using the bounding surface's derivatives instead of the named surface → fix: check which surface the question asks for.

COMMON SURFACE-AREA PATTERNS

Problem type	Minimal skeleton
Graph $z = f(x, y)$	$f_x, f_y \rightarrow \sqrt{1 + f_x^2 + f_y^2} \rightarrow$ project region → polar if circular
Sphere	parameterize by angles $\rightarrow \mathbf{X}_u \times \mathbf{X}_v$
Paraboloid above cone	solve intersection in r first; Monge formula on paraboloid

LINE INTEGRALS · DIRECT METHOD

$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$ → substitute path into $\mathbf{F}$ first
In 2D: $\int_C P dx + Q dy$
Workflow: parameterize $C \rightarrow$ compute $d\mathbf{x} \rightarrow$ write $\mathbf{F}(\mathbf{r}(t))$ explicitly $\rightarrow$ dot $\rightarrow$ integrate.

LINE-INTEGRAL PATTERNS TO DRILL

Type	Minimal symbolic template
Straight segment	$\mathbf{r}(t) = \mathbf{A} + t(\mathbf{B} - \mathbf{A})$, $0 \leq t \leq 1$
Circle	$\mathbf{r}(t) = (a \cos t, a \sin t)$ or 3D variant
Helix	$\mathbf{r}(t) = (a \cos t, a \sin t, t)$; $\mathbf{r}' = (-a \sin t, a \cos t, 1)$
Broken path	split $C = C_1 \cup C_2 \cup \dots$; integrate each, sum
Path in 2D	substitute $y = y(x)$, $dy = y'(x)dx$; single-variable integral

2023 MT2 pattern: compute work on a helix, then on a straight segment between the same endpoints, and compare. → values may differ

CIRCULATION TEMPLATE · §4.4

Circulation = $\oint_C \mathbf{F} \cdot d\mathbf{x}$ net tangential component of $\mathbf{F}$ around a closed curve. → CCW = positive

Step 1. Parameterize: circle radius $a \rightarrow \mathbf{r}(t) = (a \cos t, a \sin t)$, $0 \leq t \leq 2\pi$ → CCW

Step 2. $\mathbf{r}'(t) = (-a \sin t, a \cos t)$

Step 3. Substitute: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$ → simplify → integrate over $[0, 2\pi]$

If you see $\oint$ or "circulation" → parameterize closed curve (CCW), compute $\mathbf{r}'$, substitute $\mathbf{F}$, dot, integrate.

Common trap: wrong orientation reverses sign → fix: confirm CCW. Reversing direction flips the answer.

TOP 10 EXAM-READY DRILLS		
#	Source	Why it matters
1	2025 MT2 Q2(a)–(b)	Best complete radial-field divergence/curl drill
2	2024 MT2 Q1(a)–(c)	Fast operator identities; nearly automatic marks
3	Tutorial 7 Q1(a)–(c)	Derive radial-gradient identities, not just memorize
4	2022 MT2 Q1(b)	Best Frenet–Serret frame drill
5	2025 MT2 Q1	Best graph-surface area exam problem
6	2023 MT2 Q2(a)	Best 3D direct line-integral drill
7	2023 MT2 Q1(c)	Radial divergence with parameter condition
8	Tutorial 7 Q2	Training version of paraboloid-area setup
9	2022 MT2 Q1(c)	Curvature and torsion follow-up after frame mastery
10	Extra Problems 2 Q11(a)	Best cylindrical triple-integral setup drill

COMMON BOUND / GEOMETRY TRIGGERS

Given geometry	Immediate move
$x^2 + y^2$ everywhere	Switch to polar/cylindrical
$x^2 + y^2 + z^2$ or sphere/cone	Test spherical first
Graph $z = f(x, y)$	Project onto xy -plane; use Monge
Patch inside cylinder	Cylinder defines boundary disk only
Offset circle/cylinder	Rewrite in polar: $r =$ function of θ
First octant	All coords ≥ 0 ; angle bounds shrink
Helix	Use parametric $\hat{\mathbf{t}}$; speed is constant

COMMON MISTAKES TO ELIMINATE

Operators: writing scalar where vector is required, or vice versa. → check output type

Radial questions: failing to rewrite field in radial form before differentiating. → convert first

Frenet–Serret: using d/ds as if it were d/ds → divide by $|\mathbf{r}'|$

Surface area: forgetting projection region or using wrong surface's derivatives. → name the surface

Triple integrals: missing Jacobian r or $r^2 \sin \phi$ → write Jacobian immediately

Line integrals: not substituting both path and differential. → write $\mathbf{F}(\mathbf{r}(t))$ first

Circulation: wrong orientation reverses sign. → confirm CCW

Exam habit: before integrating, say silently: formula, bounds, Jacobian, orientation, variable order.

ONE-LINE METHOD CHECKLISTS

Operator problem: Write formula first → identify operator → differentiate componentwise → simplify.

Helix problem: Write $\mathbf{r}'$ first → $|\mathbf{r}'| \rightarrow s \rightarrow \hat{\mathbf{t}} \rightarrow \hat{\mathbf{n}} \rightarrow \hat{\mathbf{b}} \rightarrow \kappa, \tau$.

Surface-area problem: Write dS first → choose graph/parametrization → project region → integrate.

Triple-integral problem: Sketch solid first → choose coordinates → write bounds → attach Jacobian.

Line integral: Parameterize C first → compute $d\mathbf{x} \rightarrow$ substitute $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \rightarrow$ integrate.

OPERATOR TEMPLATE A · INSTANT IDENTITIES

1. $\nabla \cdot \mathbf{r} = 1 + 1 + 1 = 3$ 2. $\nabla \times \mathbf{r} = \mathbf{0}$ (all cross-partials cancel) 3. $\nabla r^2 = 2\mathbf{r}$
Derivation of 3: $\nabla(x^2 + y^2 + z^2) = (2x, 2y, 2z) = 2\mathbf{r}$ → write these three instantly for free marks
Use on: 2024 MT2 Q1(a)–(c), Tutorial 7 Q1(a), Lecture 3A operator warm-ups.

OPERATOR TEMPLATE B · RADIAL FIELD

If $\mathbf{F} = (x^2 + y^2 + z^2)^n(x, y, z)$, then
 $\mathbf{F} = r^{2n}\mathbf{r} = r^{2n+1}\hat{\mathbf{r}}$ – identify $\mathbf{F}_{\mathbf{r}} = r^{2n+1}$
Divergence route:
 $\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d}{dr}(r^2 F_r) = \frac{1}{r^2} \frac{d}{dr}(r^{2n+3}) = (2n+3)r^{2n}$
Solenoidal condition: $(2n+3)r^{2n} = 0 \Rightarrow n = -\frac{3}{2}$ (away from singularities).
Curl route: pure radial field $\Rightarrow \nabla \times \mathbf{F} = \mathbf{0}$ away from singularities. → state this directly
Common trap: losing the extra r when converting $\mathbf{r}$ to $\hat{\mathbf{r}}$ → fix: $\mathbf{r} = r\hat{\mathbf{r}}$ so $r^{2n}\mathbf{r} = r^{2n+1}\hat{\mathbf{r}}$.

OPERATOR TEMPLATE C · $\nabla(r^n)$ AND $\nabla(1/r)$

$r^n = (x^2 + y^2 + z^2)^{n/2}$
 $\partial_x(r^n) = \frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}(2x) = nxr^{n-2}$
Similarly, $\partial_y(r^n) = ny r^{n-2}$, $\partial_z(r^n) = nz r^{n-2}$ → same pattern for all
So $\nabla(r^n) = nr^{n-2}(x, y, z) = nr^{n-2}\mathbf{r}$
Set $n = -1$: $\nabla(1/r) = -r^{-3}\mathbf{r} = -\hat{\mathbf{r}}/r^2$
Use on: Tutorial 7 Q1(b)–(c), Lecture 3A Example 3.17, radial simplifications in past papers.

OPERATOR TEMPLATE D · $\nabla \times (\mathbf{k} \times \mathbf{r})$ [2023 Q1(a)]

Identity: $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$
Step 1. $\mathbf{k}$ constant $\Rightarrow \nabla \cdot \mathbf{k} = 0$, $(\mathbf{r} \cdot \nabla)\mathbf{k} = \mathbf{0}$
Step 2. $(\mathbf{k} \cdot \nabla)\mathbf{r} = \mathbf{k}$ → since $\partial_i r_j = \delta_{ij}$
Step 3. $\nabla \times (\mathbf{k} \times \mathbf{r}) = \mathbf{k}(\nabla \cdot \mathbf{r}) - \mathbf{0} + \mathbf{0} - \mathbf{k} = 3\mathbf{k} - \mathbf{k} = 2\mathbf{k}$
General rule: $\nabla \times (\mathbf{c} \times \mathbf{r}) = 2\mathbf{c}$ for ANY constant vector $\mathbf{c}$. If you see $\nabla \times (\mathbf{k} \times \mathbf{r})$ → answer is $2\mathbf{k}$ instantly.

OPERATOR TEMPLATE E · LAPLACIAN OF $\ln(r^2)$ [2023 Q1(b)]

$\ln(x^2 + y^2 + z^2) = \ln(r^2) = 2\ln r$. Set $f(r) = 2\ln r$.
Step 1. $f'(r) = 2/r$, $f''(r) = -2/r^2$
Step 2. $\nabla^2 f = f''(r) + \frac{2}{r}f'(r) = -\frac{2}{r^2} + \frac{4}{r^2} = \frac{2}{r^2}$
If you see $\nabla^2(\ln(x^2 + y^2 + z^2))$ → rewrite as $2\ln r$, apply radial Laplacian, answer is $2/r^2$.

OPERATOR TEMPLATE F · PRODUCT-RULE APPLICATION

Rules: $\nabla \cdot (u\mathbf{F}) = u\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla u$, $\nabla \times (u\mathbf{F}) = \nabla u \times \mathbf{F} + u\nabla \times \mathbf{F}$
Example: $\nabla \cdot (r^2\mathbf{r})$. Set $u = r^2$, $\mathbf{F} = \mathbf{r}$.
Step 1. $\nabla u = \nabla(r^2) = 2\mathbf{r}$; $\nabla \cdot \mathbf{r} = 3$ → instant facts
Step 2. $\nabla \cdot (r^2\mathbf{r}) = r^2(3) + \mathbf{r} \cdot (2\mathbf{r}) = 3r^2 + 2r^2 = 5r^2$
If you see $\nabla \cdot (f\mathbf{F})$ or $\nabla \times (f\mathbf{F})$ → apply product rule, then use instant identities for each piece.

HELIX TEMPLATE A · COMPLETE FS CHAIN + DARBOUX

Step 1. $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$; $\mathbf{r}'(t) = (-a \sin t, a \cos t, b)$
Step 2. $|\mathbf{r}'| = c = \sqrt{a^2 + b^2}$ → constant
Step 3. $\hat{\mathbf{t}} = \frac{1}{c}(-a \sin t, a \cos t, b)$ → normalize
Step 4. $\frac{d\hat{\mathbf{t}}}{dt} = \frac{1}{c}(-a \cos t, -a \sin t, 0)$
Step 5. $\frac{d\hat{\mathbf{t}}}{ds} = \frac{1}{c^2}(-a \cos t, -a \sin t, 0)$ → divide by c
Step 6. $\kappa = \frac{a}{c^2}$; $\hat{\mathbf{n}} = (-\cos t, -\sin t, 0)$
Step 7. $\hat{\mathbf{b}} = \hat{\mathbf{t}} \times \hat{\mathbf{n}} = \frac{1}{c}(b \sin t, -b \cos t, a)$
Step 8. Torsion: $\frac{d\hat{\mathbf{b}}}{ds} = \frac{1}{c^2}(b \cos t, b \sin t, 0) = -\tau \hat{\mathbf{n}}$ → $\tau = \frac{b}{c^2}$. Darboux: $\mathbf{\Omega} = \kappa \hat{\mathbf{b}} + \tau \hat{\mathbf{t}}$
Memorize: $\kappa = a/c^2$, $\tau = b/c^2$ where $c = \sqrt{a^2 + b^2}$. Darboux: substitute $\kappa, \tau, \hat{\mathbf{b}}, \hat{\mathbf{t}}$ from above.
Numerical check (a=3, b=4): $c = 5$, $\hat{\mathbf{t}} = \frac{1}{5}(-3 \sin t, 3 \cos t, 4)$, $\kappa = 3/25$, $\tau = 4/25$.

SURFACE-AREA TEMPLATE A · GRAPH SURFACE

If $z = f(x, y)$, then $A = \iint_R \sqrt{1 + f_x^2 + f_y^2} dA$ → Monge formula
Paraboloid: $f(x, y) = x^2 + y^2$
Step 1. $f_x = 2x$, $f_y = 2y$
Step 2. $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$ → convert to polar
Step 3. If $0 \leq z \leq 2$, then $x^2 + y^2 \leq 2$, so $r \leq \sqrt{2}$ → projection region
Step 4. $A = \int_0^{2\pi} \int_0^{\sqrt{2}} \sqrt{1 + 4r^2} r dr d\theta$
Use on: Tutorial 7 Q2 and as training for 2025 MT2 Q1.

SURFACE-AREA TEMPLATE B · 2025 CAP

Step 1. Paraboloid: $z = 2 - r^2$. Cone: $z = r$.
Step 2. Intersection: $2 - r^2 = r \Rightarrow r^2 + r - 2 = 0 \Rightarrow r = 1$ → find intersection first
Step 3. $f_x = -2x$, $f_y = -2y$ → derivatives of the paraboloid
Step 4. $\sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + 4r^2}$
Step 5. $A = \int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} r dr d\theta$
Common trap: using cone derivatives instead of paraboloid derivatives → fix: the area is of the paraboloid cap, not the cone.

LINE-INTEGRAL TEMPLATE A · HELIX

Step 1. $\mathbf{r}(t) = (\cos t, \sin t, t)$; $\mathbf{r}'(t) = (-\sin t, \cos t, 1)$
Step 2. For $\mathbf{F} = (-y, x, z)$: $\mathbf{F}(\mathbf{r}(t)) = (-\sin t, \cos t, t)$ → substitute path into $\mathbf{F}$
Step 3. $\mathbf{F} \cdot \mathbf{r}' = \sin^2 t + \cos^2 t + t = 1 + t$ → simplify!
Step 4. $W = \int_a^b (1 + t) dt$
Lesson: after substitution, ugly 3D integrals often collapse to simple scalar integrands.

LINE-INTEGRAL TEMPLATE B · STRAIGHT SEGMENT

Step 1. From A to B : $\mathbf{r}(t) = A + t(B - A)$, $0 \leq t \leq 1$
Step 2. $d\mathbf{x} = (B - A) dt$ → constant direction
Step 3. $\int_C \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot (B - A) dt$
Common trap: assuming same endpoints $\Rightarrow$ same value → fix: without conservative-field theorem (out of scope), different paths give different values.

LINE-INTEGRAL TEMPLATE C · 2D DIFFERENTIAL FORM

$\int_C P dx + Q dy$: substitute $y = y(x)$. $dy = y'(x)dx \rightarrow \int_a^b (P + Qy')dx$ → single variable
Broken path: integrate each segment separately and add. Circle: $x = a \cos t$, $y = a \sin t$ → check orientation

CIRCULATION TEMPLATE · WORKED EXAMPLE

Example: $\mathbf{F} = (-y, x)$. C : unit circle CCW.
Step 1. $\mathbf{r}(t) = (\cos t, \sin t)$. $0 \leq t \leq 2\pi$, $\mathbf{r}' = (-\sin t, \cos t)$
Step 2. $\mathbf{F}(\mathbf{r}(t)) = (-\sin t, \cos t)$
Step 3. $\mathbf{F} \cdot \mathbf{r}' = \sin^2 t + \cos^2 t = 1$
Step 4. $\oint = \int_0^{2\pi} 1 dt = 2\pi$
Lesson: after substitution, circulation integrals often simplify to constants.

TRIPLE-INTEGRAL TEMPLATE A · NESTED CARTESIAN

$\iiint f(x, y, z) dz dx dy$
Rule: integrate inside first, substitute its bounds completely, then move outward. → inside out
Never: carry the wrong variable into the next stage after evaluating bounds.

TRIPLE-INTEGRAL TEMPLATE B · CYLINDRICAL REGION

Step 1. Region: inside $x^2 + (y - 1)^2 = 1$, above $z = 0$, below $z = \sqrt{x^2 + y^2}$
Step 2. Sidewall: $x^2 + y^2 - 2y = 0 \Rightarrow r = 2 \sin \theta$ → sidewall gives r bound
Step 3. Floor $z = 0$, roof $z = r$ → floor/roof give z bounds
Step 4. $0 \leq \theta \leq \pi$, $0 \leq r \leq 2 \sin \theta$, $0 \leq z \leq r$
Step 5. $V = \int_0^\pi \int_0^{2 \sin \theta} \int_0^r r dz dr d\theta$ → the r is the Jacobian
 $\bar{z} = \frac{1}{V} \int_0^\pi \int_0^{2 \sin \theta} \int_0^r zr dz dr d\theta$ → divide by V at end
Common trap: the extra r in the integrand is the Jacobian, not from the roof $z = r$ → fix: write Jacobian separately.

TRIPLE-INTEGRAL TEMPLATE C · SPHERICAL BOUNDS

Sphere: $\mathbf{r} = a$
Cone: $z = r \Rightarrow \phi = \pi/4$
First octant: $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$
 $dV = r^2 \sin \phi dr d\phi d\theta$ → never drop $r^2 \sin \phi$
Use on: Extra Problems 2 Q8 and any sphere/cone bound-construction question.

LAST-48-HOUR PRACTICE ORDER			
Block	Problems	Purpose	
1	2024 MT2 Q1(a)–(c), 2023 MT2 Q1(a)–(b), Tut 7 Q1	Operators + cross-product + Laplacian	
2	2025 MT2 Q2(a)–(b), 2023 MT2 Q1(c)	Radial-field recognition	
3	2022 MT2 Q1(a)–(c)	Helix + FS chain	
4	Tut 7 Q2, 2025 MT2 Q1	Graph-surface area	
5	2023 MT2 Q2(a)–(b), Tut 8 Q1	Line integrals / path comparison	
6	EP2 Q8, Q11(a)	3D bounds and Jacobians	

30-SECOND PRE-SUBMISSION CHECK

1. Did I choose the correct formula/operator first?
2. Are my bounds matched to the picture and coordinate system?
3. Did I include the Jacobian r or $r^2 \sin \phi$?
4. For FS, did I convert d/dt to d/ds ?
5. For line integrals, did I substitute both the path and $d\mathbf{x}$?
6. For circulation, is my orientation correct (CCW = positive)?
7. Did I accidentally use any out-of-scope theorem shortcut?